Presentation on SMART GROW: A Low-Cost Automated Hydroponic System

Konduru Bhanu Chandrika, Mohit

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What is Hydroponic Farming?

- **Hydroponic farming** is a method of growing plants without soil, using nutrient-rich water.
- Ideal for urban areas with limited space and unsuitable land for farming.
- Traditional systems are costly, making them unsuitable for small-scale or home use.
- **SMART GROW** offers a low-cost, smart alternative with IoT integration.
- Includes sensors for:
  - **pH** level
  - **Electrical Conductivity (EC)**
  - **Water Level, Temperature**
- Data is monitored and controlled via an Android mobile application.
- System is scalable, replicable at home, and supports a variety of plants.

Why Smart Hydroponics?

- **Urban farming** faces challenges like limited land, unpredictable weather, and poor soil quality.
- Traditional hydroponic systems require constant manual monitoring and adjustments.
- **Sensors** enable real-time monitoring and automation of critical parameters.
- IoT-based control enhances precision and reduces human intervention.
- SMART GROW integrates sensors with ESP32 and mobile app for:
  - Continuous plant health monitoring
  - Automated decision making (e.g., pump control)
  - User-friendly control via smartphone
- This system ensures healthy plant growth while saving time, water, and nutrients.

Hydroponic set

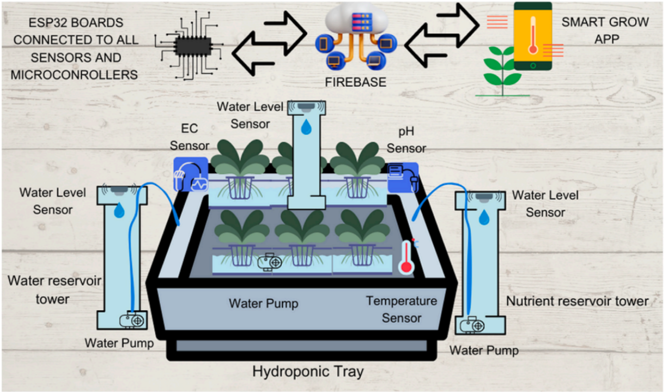


Figure: Illustration of overall Hydroponic Set

SMART GROW: System Overview

- **SMART GROW** is a DIY low-cost hydroponic system with smart monitoring and automation.
- It combines hardware and software components:
  - **Hydroponic structure:** Grow bed, water/nutrient tanks, water circulation system.
  - **Sensors:** pH, EC, temperature, and ultrasonic sensors.
  - **Microcontroller:** Dual ESP32 boards to process sensor data and control pumps.
  - **Actuators:** Water and nutrient pumps for environment regulation.
  - **Mobile App:** Android application connected via Firebase for real-time monitoring and control.
- Designed for easy replication and customization based on plant requirements.

Circuit digram of electronic components

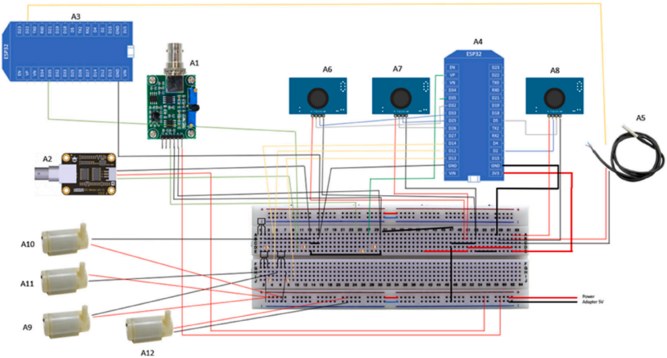


Figure:Circuit digram of electrnoic components of SMART GROW System

Sensors Used in SMART GROW

SMART GROW uses a combination of low-cost sensors to monitor critical parameters for healthy plant growth:

- **pH Sensor (PH4502C)** – Measures acidity/alkalinity of the nutrient solution.
- **EC Sensor (DFR0300)** – Measures electrical conductivity, indicating nutrient concentration.
- **Temperature Sensor (DS18B20)** – Monitors temperature of the solution.
- **Ultrasonic Sensors (JSN-SR04T)** – Three sensors to track water levels in:
  - Hydroponic tank
  - Clean water tank
  - Nutrient solution tank

Each sensor feeds real-time data to ESP32 for decision-making and automation.

How Does a pH Sensor Work?

- **pH sensors** measure the hydrogen ion (H^+) activity in a solution to determine its acidity or alkalinity.
- Most common type: **Glass electrode pH sensors**.
- Consists of two main parts:
  - **Measuring electrode:** Sensitive to hydrogen ion concentration.
  - **Reference electrode:** Provides a constant voltage.
- When immersed in solution, a **voltage is generated** based on hydrogen ion concentration.
- The voltage (mV) is converted to a pH value using the Nernst Equation:

$$E = E^0 - \frac{2.303RT}{nF} \log[H^+]$$

- A neutral solution (pH 7) gives 0 mV difference. Acidic or basic solutions cause positive or negative shifts.

Working of pH Sensor

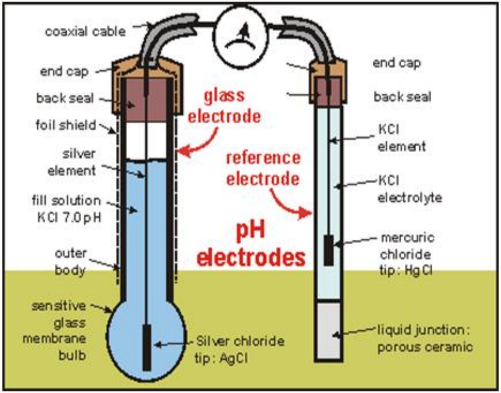
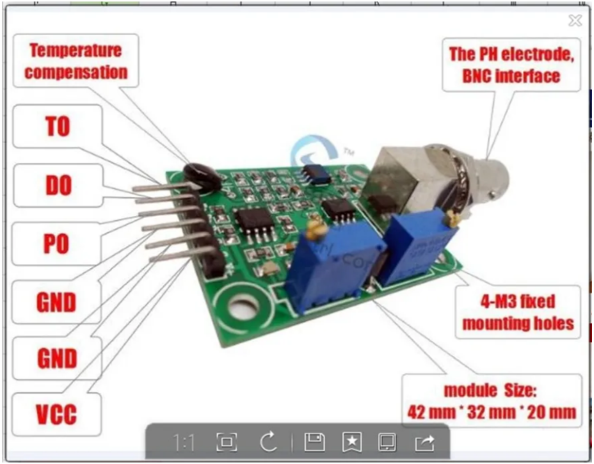


Figure: Diagram of a glass electrode and reference electrode in a pH measurement setup.

Why PH4502C Sensor for SMART GROW?

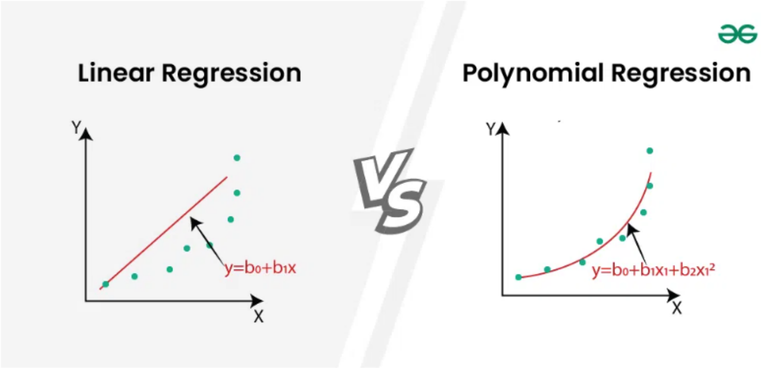
- **PH4502C** is a low-cost analog pH sensor compatible with microcontrollers.
- Features that made it suitable for SMART GROW:
  - Compact and affordable (~RM 48 / $10)
  - Wide detection range (pH 0 to 14)
  - Easily interfaces with Arduino and ESP32 via analog output
  - Comes with a signal conditioning board (amplifies and stabilizes signal)
- **Why it fits this system:**
  - Precision needed for nutrient solution monitoring
  - Easily integrated into DIY builds
  - Supports continuous data acquisition
- Alternative industrial pH sensors are more expensive and bulky, which is not ideal for compact home systems.

The PH4502C sensor pin out



Calibration of PH4502C Sensor

- **Issue:** PH4502C outputs 0–5V, but ESP32 can only read up to 3.3V analog input.
- **Solution:** Used a voltage divider with two 1kΩ resistors to scale down voltage safely.
- **Calibration Process:**
  - pH sensor readings were taken at known pH buffer solutions (4.0, 7.0, 10.0).
  - Sensor output voltages were recorded.
  - **Polynomial regression** was used to fit a curve mapping voltage to actual pH.
  - Calibration equation was integrated into Arduino code.
- Sensor values are sent to Firebase via ESP32.
- Once calibrated, the sensor requires no further adjustment for normal operation.


Linear Regression
Y
X
$y=b_0+b_1x$
VS
Polynomial Regression
Y
X
$y=b_0+b_1x_1+b_2x_1^2$

Placement & Role of pH Sensor in SMART GROW

- pH sensor is immersed directly into the hydroponic nutrient reservoir.
- Connected to ESP32 (Board A4), along with ultrasonic sensors.
- ESP32 continuously reads pH value and updates Firebase in real-time.
- **If pH is too high:**
  - Water pump 2 adds clean water to dilute the solution.
- **If pH is too low:**
  - Water pump 3 adds nutrient solution to correct acidity.
- Sensor ensures plants always have optimal pH for nutrient uptake.

How Does an Ultrasonic Sensor Work?

- **Ultrasonic sensors** measure distance using high-frequency sound waves.
- The sensor emits a pulse and waits for the echo to return after reflecting from a surface.
- Time delay is measured and distance is calculated as:

$$\text{Distance} = \frac{(\text{Time} \times \text{Speed of Sound})}{2}$$

- Accurate for detecting liquid levels and solid objects.
- Suitable for use in water tanks due to non-contact nature.
- Ideal for automation and real-time monitoring in hydroponic systems.

Working of Ultrasonic Sensor

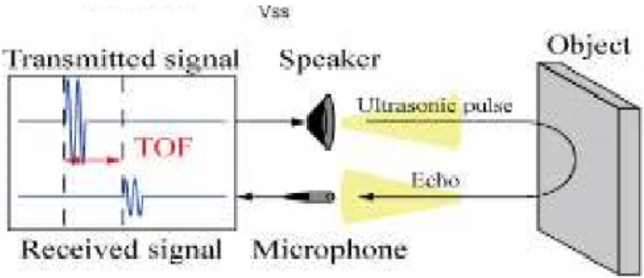
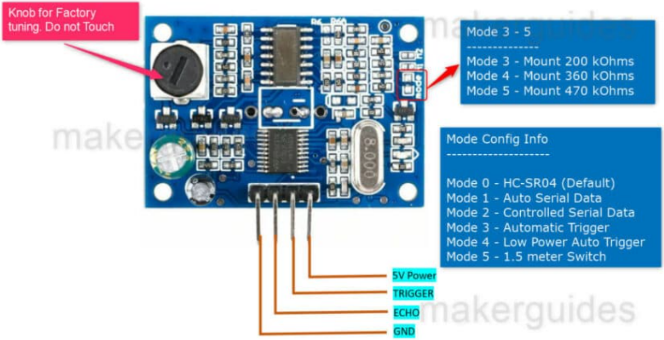


Figure: ultrasonic sensor emitting and receiving sound waves to measure water level.

Why JSN-SR04T for SMART GROW?

- **JSN-SR04T** is a waterproof ultrasonic distance sensor.
- Key reasons for selection:
  - Resistant to humidity and water splashes — ideal for hydroponic environments.
  - Operates in 20–400 cm range with ˜2 mm accuracy.
  - Simple 4-pin interface (VCC, Trig, Echo, GND).
  - Inexpensive and reliable for DIY integration.
- Readily compatible with Arduino and ESP32 boards.
- Ensures stable water level monitoring without sensor degradation.

The JSN-SR04T sensor pin out



Ultrasonic Sensor: Calibration and Usage

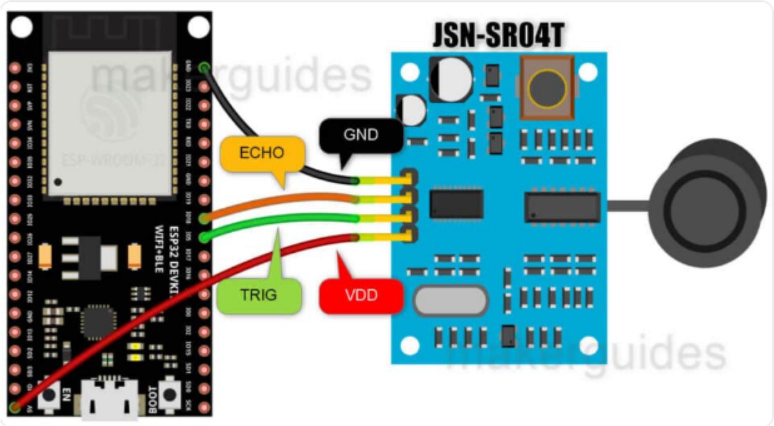
- JSN-SR04T outputs a digital pulse length representing distance.
- ESP32 is programmed to:
  - Send a HIGH pulse to `Trig` pin for 10 µs.
  - Measure duration of pulse on `Echo` pin.
- Distance is converted to water level percentage.
- **Water Level Percentage Formula:**

$$\text{Water Level \%} = \left(1 - \frac{\text{distance}}{\text{max\_distance}}\right) \times 100$$

where:

  - `distance` = measured distance from the Echo pin (in cm)
  - `max_distance` = maximum distance the sensor can measure (in cm)
- **Challenge:** Interference from other components (pH/EC sensors).
- **Solution:** Used separate ESP32 board and placed ultrasonic sensors higher to avoid reflection noise.
- Data is uploaded to Firebase and visualized via app.

The JSN-SR04T sensor hardware connection



Ultrasonic Sensors in SMART GROW

- Total of **3 JSN-SR04T sensors** used:
  - Hydroponic tank water level
  - Clean water tank level
  - Nutrient solution tank level
- Mounted on top of each tank facing downward.
- Data is used to:
  - Trigger pumps when level is low or high
  - Alert user via app if refill is needed
  - Maintain balance in nutrient concentration
- Ensures system is autonomous and water-efficient.

DS18B20 Temperature Sensor

- The DS18B20 is a digital temperature sensor using the **1-Wire protocol**.
- Communicates over a single data line (DQ) with a pull-up resistor.
- Provides **9 to 12-bit** temperature readings in digital form.
- Can operate with **external power** or in **parasite power mode**.
- Contains internal components like:
  - Temperature sensor
  - Scratchpad memory
  - Alarm trigger registers (TH and TL)
  - Unique 64-bit ROM ID
  - CRC generator for data integrity
- Suitable for multiple sensors on the same bus.

Working of DS18B20 Temp sensor

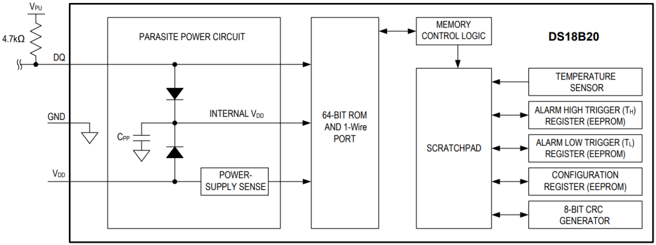


Figure: Block Diagram.

DS18B20 Temp sensor

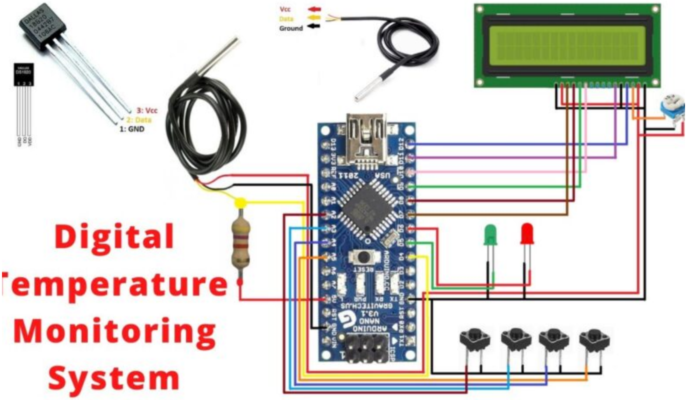


Figure: Circuit Diagram

EC Sensor

EC Sensor

- Electrical Conductivity (EC) sensors measure the concentration of dissolved ions (nutrients) in water.
- The DFRO300 EC sensor is widely used in agriculture, hydroponics, and water quality monitoring.
- Real-time EC monitoring helps ensure plants receive optimal nutrients.

Working Principle

- Alternating voltage is applied across two electrodes in the solution.
- Ionic current is measured and used to calculate resistance.
- Conductivity is calculated using:

$$EC = \frac{1}{R} \cdot K$$

  where K is the cell constant.
- Temperature compensation is applied for accuracy.

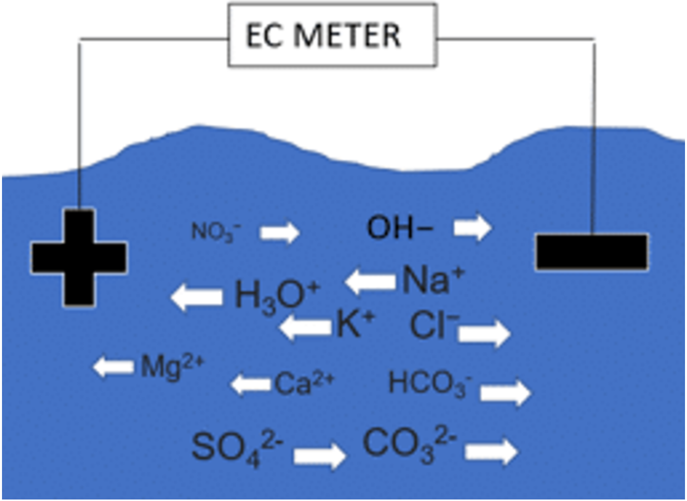
System Behavior Based on EC Levels

- **If EC is high:**
  - Nutrient concentration is too high.
  - System dilutes solution with fresh water.
- **If EC is low:**
  - Nutrient concentration is too low.
  - System adds nutrients automatically.

IoT-Based Automation

- EC sensor connected to a microcontroller (e.g., Arduino, ESP32).
- Real-time data is sent to a cloud server or mobile app.
- Automated dosing pumps or valves are triggered based on EC readings.

System Diagram



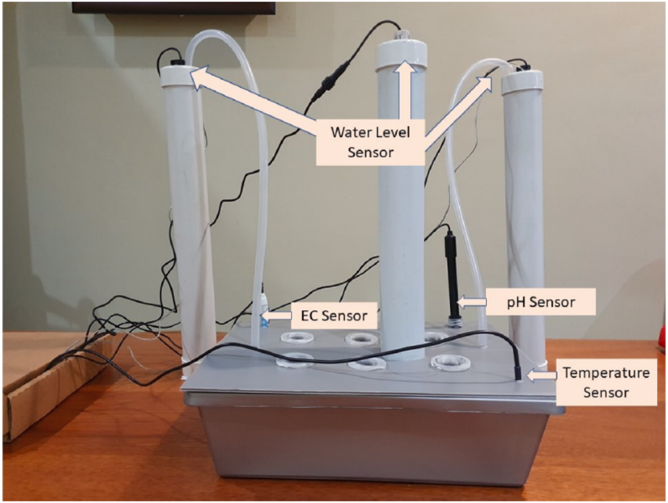
Conclusion

- EC sensors like DFRO300 enable precise nutrient monitoring.
- Automated control through IoT enhances crop yield and resource efficiency.
- A smart, scalable solution for agriculture and hydroponics.

Micro Water Pumps in SMART GROW

- Total of **4 micro water pumps** are used, each with a specific automated function:
  - **Pump 1:** Removes excess water from the hydroponic tank.
  - **Pump 2:** Adds clean water when pH or EC value is too high.
  - **Pump 3:** Adds nutrient solution when pH or EC is too low.
  - **Pump 4:** Circulates nutrient solution within the tank.
- Pump control is triggered based on sensor input values received by ESP32.
- Pumps are driven via BC639 transistors acting as switches.
- Enables autonomous correction of pH and nutrient concentration without manual intervention.

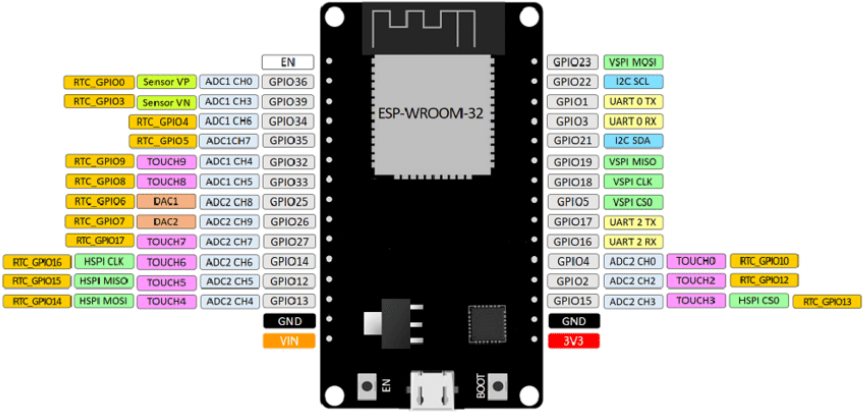
The compelte hydroponic setup with sensors



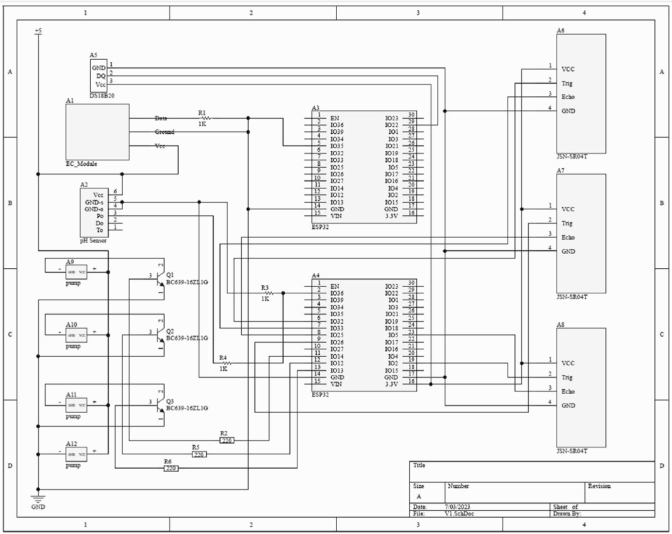
Sensor Integration with ESP32

- **ESP32** microcontroller is the brain of SMART GROW.
- Two DOIT ESP32 DEVKIT V1 boards are used:
  - **ESP32 A3:** Handles EC sensor data exclusively.
  - **ESP32 A4:** Manages pH and ultrasonic sensors and controls water pumps.
- **Why two ESP32s?**
  - Avoids sensor interference and unstable readings.
  - Each board operates on a dedicated circuit for better performance.
- Data is transmitted to a Firebase real-time database.
- Calibration and signal conditioning are performed before uploading values.
- Enables real-time feedback, automation, and remote user interaction via mobile app.

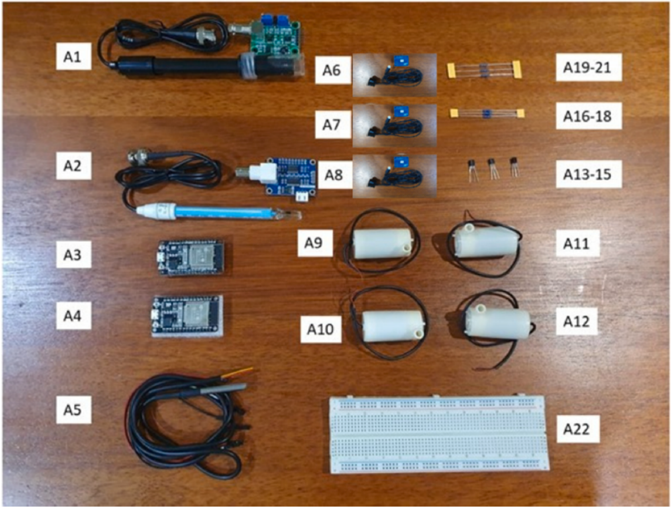
ESP32 microcontroller pin out



The Overall integration of ESP32



The Overall parts and their description



The Overall parts and fucntions

Table 2
Overview of Parts and Description of their Function.

Label	Part Name	Description
A1	pH circuit board and pH Probe	Detect pH level of the solution in the hydroponic reservoir
A2	EC circuit board and EC Probe	Detect EC level of the solution in the hydroponic reservoir
A3	ESP32 Microcontroller	Send and receive data from EC sensor to Firebase
A4	ESP32 Microcontroller	Manage all module except EC sensor including send and receive data from sensors to Firebase and environment regulation
A5	Temperature Sensor	Detect temperature of solution in the hydroponic tank
A6	Ultrasonic Sensor 1	Detect the water level in the hydroponic tank
A7	Ultrasonic Sensor 2	Detect the water level in the clean water tank
A8	Ultrasonic Sensor 3	Detect the water level in the nutrient solution tank
A9	Water Pump 1	Pump water out of hydroponic tank if the water level is too high
A10	Water Pump 2	Pump clean water into hydroponic tank if the pH or EC value is too high
A11	Water Pump 3	Pump nutrient solution into hydroponic tank if the pH or EC value is too low
A12	Water Pump 4	Pump that circulates the flow of solution inside the hydroponic tank
A13	Transistor BC639 1	To act as switch for Water Pump 2
A14	Transistor BC639 2	To act as switch for Water Pump 3
A15	Transistor BC639 3	To act as switch for Water Pump 1
A16	Resistor 220 Ohm 1	To act as base resistor for Transistor 1
A17	Resistor 220 Ohm 2	To act as base resistor for Transistor 2
A18	Resistor 220 Ohm 3	To act as base resistor for Transistor 3
A19	Resistor 1000 Ohm 1	To act as voltage divider for EC Sensor
A20	Resistor 1000 Ohm 2	To act as voltage divider for pH Sensor
A21	Resistor 1000 Ohm 3	To act as voltage divider for Temperature Sensor
A22	MB102 Bread Board	For wiring electrical circuits

Data Flow & Mobile App Interface

- SMART GROW uses the **Firebase Realtime Database** for storing and accessing sensor data.
- **Data Flow:**
  - Sensors send data to ESP32 boards.
  - ESP32 uploads data to Firebase.
  - Mobile app fetches data and updates interface.
- **Mobile App Features:**
  - View current sensor readings.
  - Customize threshold values for pH, EC, and water level.
  - Receive notifications if values go out of range.
  - Add/update/delete plant information and notes.
- Designed to be **user-friendly, responsive**, and ideal for daily monitoring.

Overall Processes

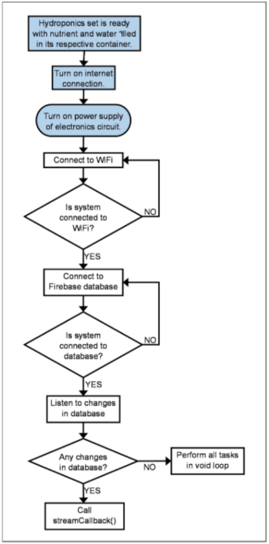


Figure: Overall Processes

Automated Processes Repeatedly Carried Out Over Time

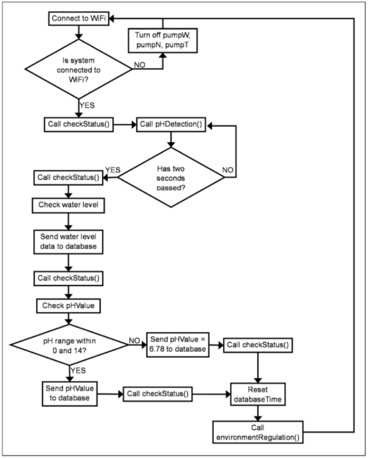
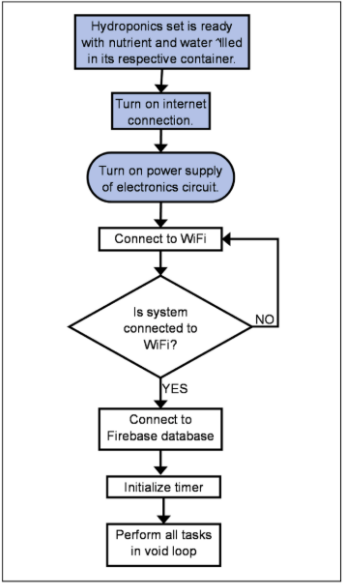


Figure : Automated Processes Repeatedly Carried Out Over Time

Overall Processes Occurring



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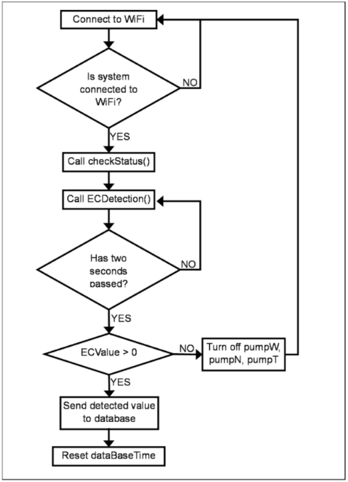
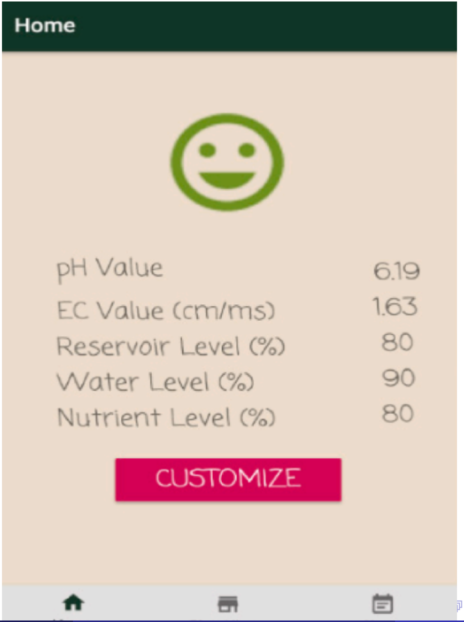


Figure:Automated Processes Repeatedly Carried Out Over Time

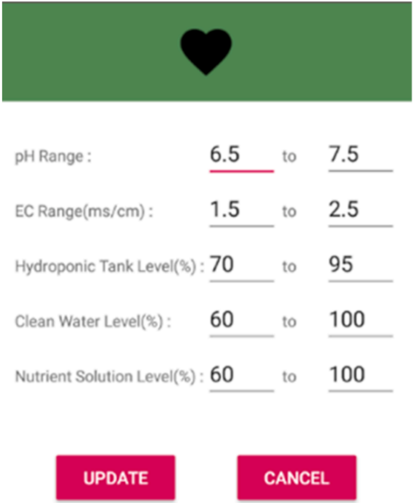
Testing and Results

- **Integration Testing:**
  - Initial issues occurred when multiple analog sensors were connected to the same ESP32.
  - Solved by using separate ESP32 boards for EC and pH sensors.
  - Proper placement of ultrasonic sensors improved reading accuracy.
- **Functional Testing:**
  - All app features (data view, plant library, alerts) tested and worked as expected.
  - Automatic pump actions matched threshold logic set in the app.
- **User Testing:**
  - Conducted with real plant growth—Ensabi (mustard plant).
  - Observed better growth and more leaves in SMART GROW compared to soil-based method.
  - App rated well for usability, though minor feedback on interactivity and documentation.
- **Conclusion:** SMART GROW offers a reliable and efficient alternative to traditional hydroponics with low cost and smart automation.

Functional Testing



Functional Testing



Testing and Results

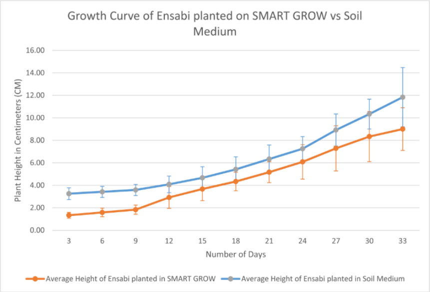
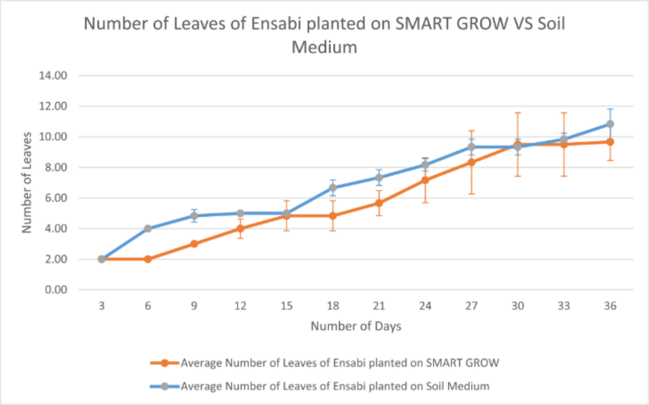


Figure: Number of Leaves of Ensabi planted on SMART GROW vs Soil Medium

Testing and Results



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